

Companion to Bird & de Moor, Algebra of Programming

§6. Recursive Programs

§6.1. Digits of a number

(6.1a)

$$([g,h])^\circ = (g^\circ \text{ wrap}) \cup (h^\circ \neg \text{Digit}([g,h])^\circ) \text{ snoc}$$

the converse of a fold over decimals undoes the algebra $[g,h]$ and then either **wrap**, or the fold's converse on the front and **snoc**

1	$[g,h]^\circ$	$(\emptyset) = [\text{wrap}, \text{snoc}]^\circ \text{ Digit}^{+-}(\neg \text{Digit}(\emptyset)) \emptyset$
2	$= ([\text{wrap}, \text{snoc}]^\circ \text{ Digit}^{+-}(\neg \text{Digit}([g,h])^\circ)) [g,h]^\circ$	$(R + S) [P,Q] = [R P, S Q]$
3	$= ([\text{wrap}, \text{snoc}]^\circ [g, \neg \text{Digit}([g,h])^\circ] h)^\circ$	$[U,V]^\circ [R,S] = (U^\circ R) \cup (V^\circ S)$
4	$= ((\text{wrap}^\circ g) \cup (\text{snoc}^\circ \neg \text{Digit}([g,h])^\circ) h)^\circ$	$(R \cup S)^\circ = S^\circ \cup R^\circ$
5	$= (\text{snoc}^\circ \neg \text{Digit}([g,h])^\circ) h^\circ \cup (\text{wrap}^\circ g)^\circ$	$R \cup S = S \cup R$
6	$= (\text{wrap}^\circ g)^\circ \cup (\text{snoc}^\circ \neg \text{Digit}([g,h])^\circ) h^\circ$	$(R S)^\circ = S^\circ R^\circ$
7	$= (g^\circ \text{ wrap}) \cup ((\neg \text{Digit}([g,h])^\circ) h^\circ \text{ snoc})$	$(R S)^\circ = S^\circ R^\circ$
8	$= (g^\circ \text{ wrap}) \cup (h^\circ \neg \text{Digit}([g,h])^\circ) \text{ snoc}$	$\neg \text{Digit}(R)^\circ = \neg \text{Digit}(R^\circ)$
9	$= (g^\circ \text{ wrap}) \cup (h^\circ \neg \text{Digit}([g,h])^\circ) \text{ snoc}$	

$\text{val}^\circ = (\text{embed}^\circ \text{ wrap}) \cup (\text{op}^\circ \neg \text{Digit}(\text{val}^\circ) \text{ snoc})$

a number is read back into digits either as one nonzero digit, or by splitting off its last digit and reading back the rest

(6.1b)

1	$\text{op}^\circ(m,p) \iff \pi_1(p) = m/10 \wedge \pi_2(p) = m \% 10$ op° splits a number into its quotient and remainder by 10
2	$(\exists p. 0 < \pi_1 p \wedge \text{op}^\circ(m,p)) \iff 10 \leq m$ op° gives a pair with a nonzero first component exactly at the numbers with two or more digits
3	$(\exists d. \text{embed}^\circ(m,d)) \iff 0 < m \wedge m < 10$ embed° gives a digit exactly at the one-digit numbers

§6.2. Least fixed points

(6.2a)

$\text{Monotonic}(\varphi) \triangleq \forall X. \forall Y. X \sqsubseteq Y \implies \varphi(X) \sqsubseteq \varphi(Y)$ — φ maps the hom-set $A \rightarrow B$ into itself with $X \sqsubseteq Y \implies \varphi(X) \sqsubseteq \varphi(Y)$.

$(\mu X : \varphi(X))$ is the least $X : A \rightarrow B$ with $\varphi(X) \sqsubseteq X$.

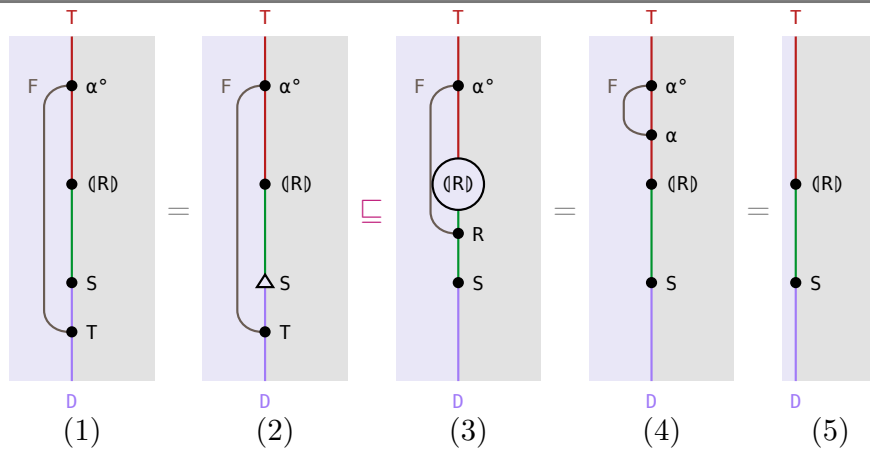
(6.2b)

the law	what it says
1 $\varphi(Y) \sqsubseteq Y \implies (\mu X : \varphi(X)) \sqsubseteq Y$	to bound $(\mu X : \varphi(X))$ above, exhibit one Y the body does not grow past — the half §Section 6.3 and every chapter after it uses
2 $\text{Monotonic}(\varphi) \implies \varphi(\mu X : \varphi(X)) = (\mu X : \varphi(X))$	Knaster–Tarski: the least solution of $\varphi(X) \sqsubseteq X$ already solves $\varphi(X) = X$, so the least prefix point and the least fixed point are one relation
3 $\alpha^\circ F(X) R \sqsubseteq X \implies \langle R \rangle \sqsubseteq X$ (6.2)	a relation X that contains one unfolding of the fold's recursion at X contains the fold
4 $X \sqsubseteq \alpha^\circ F(X) R \implies X \sqsubseteq \langle R \rangle$ (6.3)	a relation X contained in one unfolding of the fold's recursion at X is contained in the fold

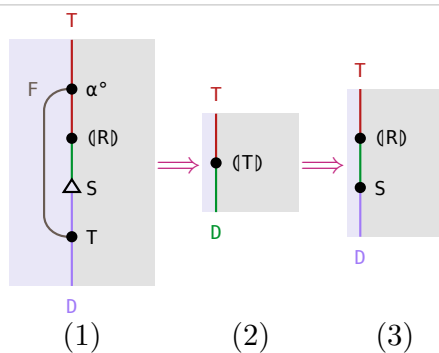
$$F(S) \ T \sqsubseteq R \ S \implies \langle T \rangle \sqsubseteq \langle R \rangle \ S$$

(6.2c)

if S followed by R absorbs T after $F(S)$, the fold of T is below the fold of R followed by S



1 $\alpha^\circ F(\langle R \rangle S) T$	
2 $= \alpha^\circ F(\langle R \rangle) F(S) T$	$F(S) \ T \sqsubseteq R \ S$
3 $\sqsubseteq \alpha^\circ F(\langle R \rangle) R \ S$	$\alpha \ \langle R \rangle = F(\langle R \rangle) \ R$
4 $= \alpha^\circ \alpha \ \langle R \rangle \ S$	$\alpha^\circ \alpha = 1$
5 $= \langle R \rangle \ S$	

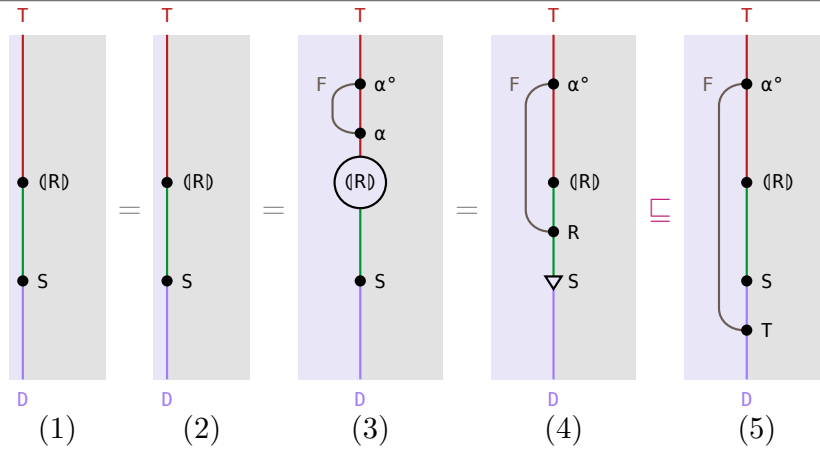


1 $F(S) \ T \sqsubseteq R \ S$	$F(S) \ T \sqsubseteq R \ S$ $\implies \alpha^\circ F(\langle R \rangle S) T \sqsubseteq \langle R \rangle \ S$
2 $\implies \alpha^\circ F(\langle R \rangle S) T \sqsubseteq \langle R \rangle \ S$	$\alpha^\circ F(X) R \sqsubseteq X$ $\implies \langle R \rangle \sqsubseteq X$
3 $\implies \langle T \rangle \sqsubseteq \langle R \rangle \ S$	

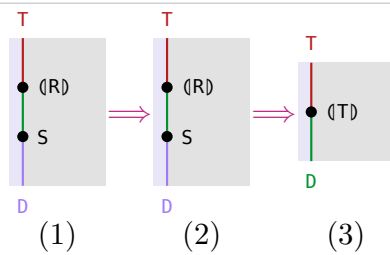
$$R \ S \sqsubseteq F(S) \ T \Rightarrow \langle R \rangle \ S \sqsubseteq \langle T \rangle$$

(6.2d)

if R followed by S is absorbed by $F(S)$ followed by T , the fold of R followed by S is below the fold of T



1	$\langle R \rangle \ S$	
2	$= \langle R \rangle \ S$	$\alpha^\circ \ \alpha = 1$
3	$= \alpha^\circ \ \alpha \ \langle R \rangle \ S$	$\alpha \ \langle R \rangle = F(\langle R \rangle) \ R$
4	$= \alpha^\circ \ F(\langle R \rangle) \ R \ S$	$R \ S \sqsubseteq F(S) \ T$
5	$\sqsubseteq \alpha^\circ \ F(\langle R \rangle \ S) \ T$	



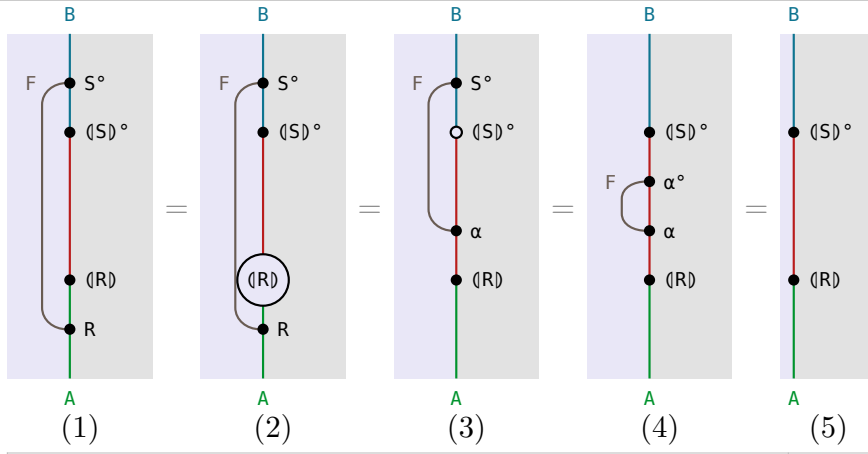
1	$R \ S \sqsubseteq F(S) \ T$	$R \ S \sqsubseteq F(S) \ T$ $\Rightarrow \langle R \rangle \ S \sqsubseteq \alpha^\circ \ F(\langle R \rangle \ S) \ T$
2	$\Rightarrow \langle R \rangle \ S \sqsubseteq \alpha^\circ \ F(\langle R \rangle \ S) \ T$	$X \sqsubseteq \alpha^\circ \ F(X) \ R$ $\Rightarrow X \sqsubseteq \langle R \rangle$
3	$\Rightarrow \langle R \rangle \ S \sqsubseteq \langle T \rangle$	

§6.3. Hylomorphisms

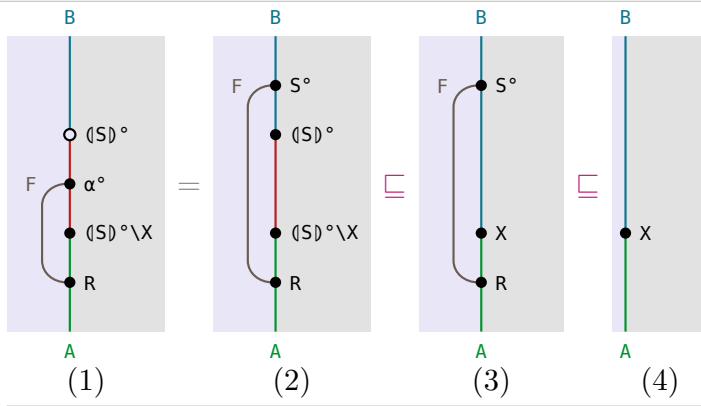
$$(S)^\circ (R) = (\mu X : S^\circ F(X) R)$$

(6.3a)

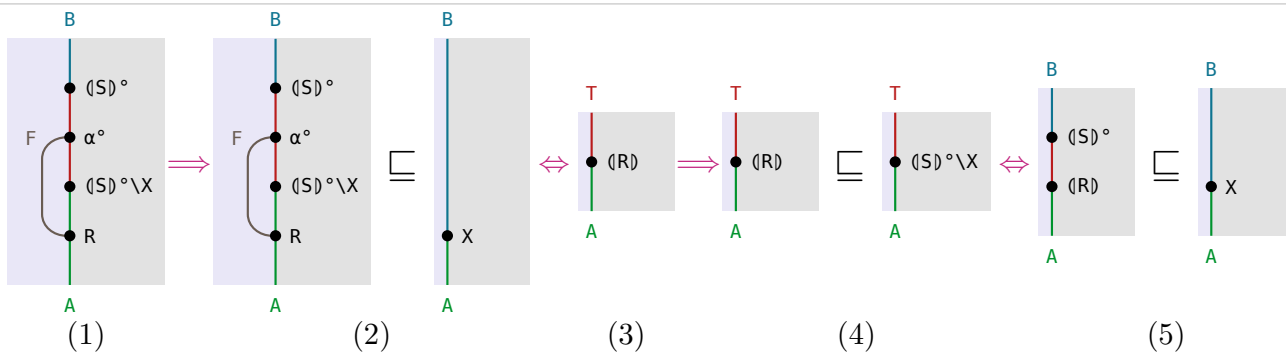
hylomorphism theorem: a hylomorphism is the least fixed point of a certain recursion equation



1	$S^\circ F((S)^\circ (R)) R$	
2	$= S^\circ F((S)^\circ) F((R)) R$	$\alpha (R) = F((R)) R$
3	$= S^\circ F((S)^\circ) \alpha (R)$	$(R)^\circ \alpha^\circ = R^\circ F((R)^\circ)$
4	$= (S)^\circ \alpha^\circ \alpha (R)$	$\alpha^\circ \alpha = 1$
5	$= (S)^\circ (R)$	



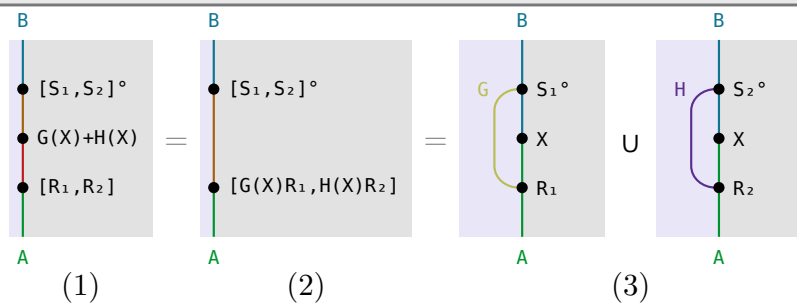
1	$(S)^\circ \alpha^\circ F((S)^\circ \setminus X) R$	$(R)^\circ \alpha^\circ = R^\circ F((R)^\circ)$
2	$= S^\circ F((S)^\circ ((S)^\circ \setminus X)) R$	$S (S \setminus R) \subseteq R$
3	$\subseteq S^\circ F(X) R$	$S^\circ F(X) R \subseteq X$
4	$\subseteq X$	



1	$S^\circ F(X) R \subseteq X$	$S^\circ F(X) R \subseteq X$
2	$\Rightarrow (S)^\circ \alpha^\circ F((S)^\circ \setminus X) R \subseteq X$	$\Rightarrow (S)^\circ \alpha^\circ F((S)^\circ \setminus X) R \subseteq X$
3	$\Leftrightarrow \alpha^\circ F((S)^\circ \setminus X) R \subseteq (S)^\circ \setminus X$	$X \subseteq S \setminus R \Leftrightarrow S X \subseteq R$
4	$\Rightarrow (R) \subseteq (S)^\circ \setminus X$	$\alpha^\circ F(X) R \subseteq X$
5	$\Leftrightarrow (S)^\circ (R) \subseteq X$	$\Rightarrow (R) \subseteq X$
		$X \subseteq S \setminus R \Leftrightarrow S X \subseteq R$

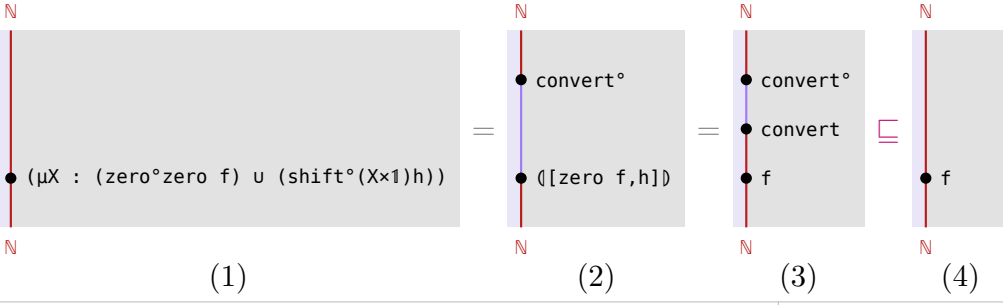
$$([S_1, S_2]^\circ)^\circ ([R_1, R_2])^\circ = (\mu X : (S_1^\circ G(X) R_1) \cup (S_2^\circ H(X) R_2)) \quad (6.3b)$$

when both algebras are case splits over the same sum, the recursion runs each case on its own and unites the results



1	$[S_1, S_2]^\circ (G(X) + H(X)) [R_1, R_2]$	$(R + S) [P, Q] = [R P, S Q]$
2	$= [S_1, S_2]^\circ [G(X) R_1, H(X) R_2]$	$[U, V]^\circ [R, S] = (U^\circ R) \cup (V^\circ S)$
3	$= (S_1^\circ G(X) R_1) \cup (S_2^\circ H(X) R_2)$	

§6.4. Fast exponentiation and modulus computation

$\text{shift } f = (f \times 1) \ h \Rightarrow (\mu X : (\text{zero}^\circ \text{ zero } f) \cup (\text{shift}^\circ (X \times 1) h)) \sqsubseteq f$ once f turns shift into h, the recursion that halves the argument at each step computes f		(6.4a)
 (1) (2) (3) (4)		
1	$(\mu X : (\text{zero}^\circ \text{ zero } f) \cup (\text{shift}^\circ (X \times 1) h))$	$\text{convert}^\circ ([g, h]) = (\mu X : (\text{zero}^\circ g) \cup (\text{shift}^\circ (X \times 1) h))$
2	$= \text{convert}^\circ ([\text{zero } f, h])$	$\text{zero } f = g \wedge \text{shift } f = (f \times 1) \ h$
3	$= \text{convert}^\circ \text{ convert } f$	$\Rightarrow \text{convert } f = ([g, h])$
4	$\sqsubseteq f$	$\text{Simple}(\text{convert})$

$(\mu X : (\text{zero}^\circ \text{ one}) \cup (\text{shift}^\circ (X \times 1) \text{ op}(a))) \sqsubseteq \text{exp}(a)$ the recursion that halves the exponent at each step computes a to the power b		(6.4b)
$\text{zero exp}(a) = \text{one}$ a to the power 0 is 1		
$\text{shift exp}(a) = (\text{exp}(a) \times 1) \text{ op}(a)$ a to the power $2n+d$ is $\text{op}(a)$ of a to the power n and d		

$(\mu X : (\text{zero}^\circ \text{ zero}) \cup (\text{shift}^\circ (X \times 1) \text{ op}(b))) \sqsubseteq \text{mod}(b)$ the recursion that halves a at each step computes the remainder of a by b		(6.4c)
$\text{zero mod}(b) = \text{zero}$ $0 \text{ mod } b$ is 0		
$\text{shift mod}(b) = (\text{mod}(b) \times 1) \text{ op}(b)$ $(2a+d) \text{ mod } b$ is $\text{op}(b)$ of $a \text{ mod } b$ and d		

§6.5. Unique fixed points

(6.5a)

$$R R \sqsubseteq S R \Rightarrow (X / R / S) R R \sqsubseteq X$$

((X/R)/S) followed by RR stays below X when $RR \sqsubseteq SR$

$$\text{inductive}(S) \wedge R R \sqsubseteq S R \Rightarrow \text{inductive}(R)$$

R is inductive when $RR \sqsubseteq SR$ for an inductive S

1	$(X / R / S) R R$	$R R \sqsubseteq S R$
2	$\sqsubseteq (X / R / S) S R$	$(R / S) S \sqsubseteq R$
3	$\sqsubseteq (X / R) R$	$(R / S) S \sqsubseteq R$
4	$\sqsubseteq X$	

§6.5.1. Membership

(6.5.1a)

member = $\emptyset$

a constant relator records no elements, so its membership is empty

member = [member, member]

a member of $F+G$ is a member of whichever summand the value is in

member = $(\pi_1 \text{ member}) \cup (\pi_2 \text{ member})$

a member of $F \times G$ is a member of either component

§6.5.2. Consequences

(6.5.2a)

$$\text{inductive}(S \text{ member}) \wedge X = S F(X) R \wedge Y = S F(Y) R \Rightarrow X = Y$$

when $S \text{ member}$ is inductive, two solutions X, Y of $X = SF(X)R$ are equal

(6.5.2b)

$$\text{inductive}(S \text{ member}) \wedge \text{Entire}(S) \wedge \text{Entire}(R) \wedge S F(X) R \sqsubseteq X \Rightarrow \text{Entire}(X)$$

when $S \text{ member}$ is inductive and S, R are entire, a pre-fixed point of $X \mapsto SF(X)R$ is entire

(6.5.2c)

$$\text{inductive}(g \text{ member}) \wedge \text{Map}(g) \wedge \text{Map}(f) \wedge X = g F(X) f \Rightarrow \text{Map}(X)$$

when $g \text{ member}$ is inductive and f, g are maps, the solution of $X = gF(X)f$ is a map

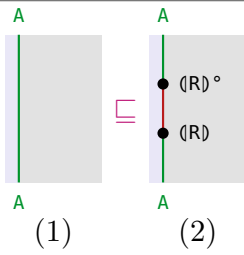
$$X = g F(X) f \wedge \text{Simple}(g) \wedge \text{Simple}(f) \wedge Y^\circ X \sqsubseteq 1 \Rightarrow (g F(Y) f)^\circ X \sqsubseteq 1$$

$(gF(Y)f)^\circ$ followed by X stays below 1 when g, f are simple and $Y^\circ X \sqsubseteq 1$

1	$(g F(Y) f)^\circ X$	$X = g F(X) f$
2	$= (g F(Y) f)^\circ g F(X) f$	$(R S)^\circ = S^\circ R^\circ$
3	$= f^\circ F(Y)^\circ g^\circ g F(X) f$	PreservesRecip(F)
4	$= f^\circ F(Y)^\circ g^\circ g F(X) f$	Simple(g)
5	$\sqsubseteq f^\circ F(Y)^\circ X f$	$Y^\circ X \sqsubseteq 1$
6	$\sqsubseteq f^\circ f$	Simple(f)
7	$\sqsubseteq 1$	

$$\text{inductive}(R^\circ \text{ member}) \wedge 1 \sqsubseteq R^\circ R \Rightarrow 1 \sqsubseteq (R)^\circ (R) \quad (6.5.2d)$$

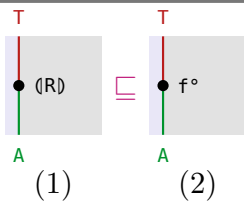
when $R^\circ \text{member}$ is inductive and R is surjective, the fold (R) is surjective



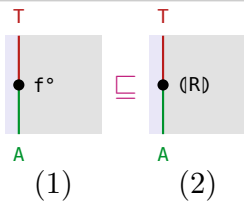
1	1	$\text{inductive}(S \text{ member}) \wedge \text{Entire}(S) \wedge \text{Entire}(R) \wedge S$ $F(X) R \sqsubseteq X$ $\Rightarrow \text{Entire}(X) \text{ at } S=R^\circ$
2	$\sqsubseteq (R)^\circ (R)$	

$$\text{Map}(f) \wedge R f \sqsubseteq F(f) \alpha \wedge 1 \sqsubseteq R^\circ R \Rightarrow f^\circ = (R) \quad (6.5.2e)$$

a map f with $Rf \sqsubseteq F(f)\alpha$, for a surjective R , has the fold (R) as its converse



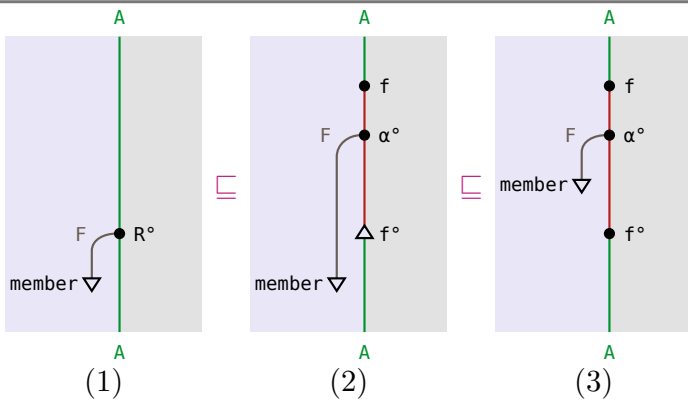
1	(R)	shunting f , fusion, $Rf \sqsubseteq F(f)\alpha$
2	$\sqsubseteq f^\circ$	



1	f°	$\text{inductive}(R^\circ \text{ member}) \wedge 1 \sqsubseteq R^\circ R$ $\Rightarrow 1 \sqsubseteq (R)^\circ (R) \text{ and } (R)f \sqsubseteq 1$
2	$\sqsubseteq (R)$	

$$\text{Map}(f) \wedge R f \sqsubseteq F(f) \alpha \Rightarrow R^\circ \text{ member} \sqsubseteq f \alpha^\circ \text{ member } f^\circ \quad (6.5.2f)$$

$R^\circ \text{member}$ is below $\alpha^\circ \text{member}$ conjugated by f , so it is inductive when $\alpha^\circ \text{member}$ is



1	$R^\circ \text{ member}$	$\text{Map}(f) \wedge R f \sqsubseteq F(f) \alpha$ $\Rightarrow R^\circ \sqsubseteq f \alpha^\circ F(f^\circ)$
2	$\sqsubseteq f \alpha^\circ F(f^\circ) \text{ member}$	$F(R) \text{ member} \sqsubseteq \text{member } R$
3	$\sqsubseteq f \alpha^\circ \text{ member } f^\circ$	

§6.6. Sorting by selection

§6.6.1. The function sort

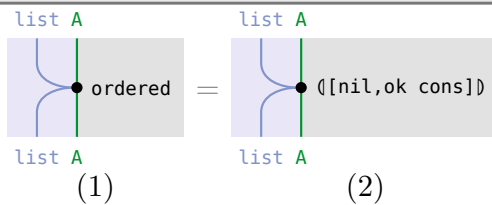
(6.6.1a)

	name	type	definition	meaning
1	orderedP(R)	$P[A]$	$\text{orderedP}[] = \text{true}, \text{orderedP}([a] \# xs) = (\forall x. x \in xs \Rightarrow R(a, x)) \wedge \text{orderedP}(xs)$	every element is R -below each element after it
2	ordered	$[A] \rightarrow [A]$	$xs \text{ ordered } ys \iff xs = ys \wedge \text{orderedP}(xs)$	the coreflexive passing exactly the lists orderedP holds of
3	ok	$A \times [A] \rightarrow A \times [A]$	$(a, xs) \text{ ok } q \iff (a, xs) = q \wedge (\forall x. x \in xs \Rightarrow R(a, x))$	the coreflexive passing (a, x) when a is R -below every element of x

ordered = $\llbracket \text{nil}, \text{ok cons} \rrbracket$

(6.6.1b)

a list is sorted exactly when the fold that rebuilds it passes **ok** at every **cons**, i.e. each head is **R**-below every element after it



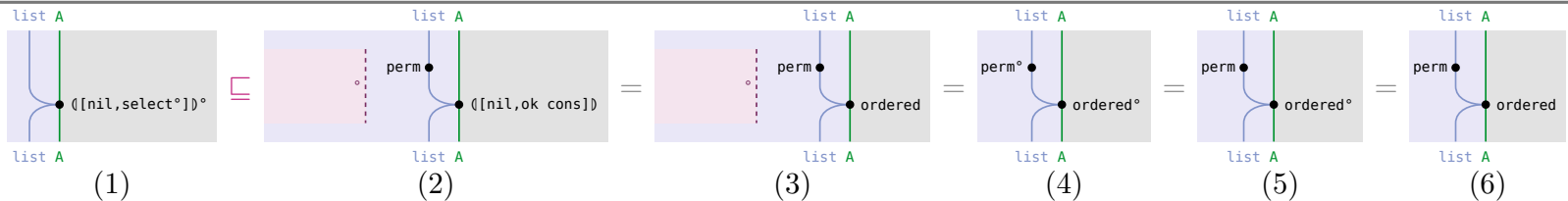
1	ordered	fold uniqueness
2	= $\llbracket \text{nil}, \text{ok cons} \rrbracket$	

§6.6.2. Selection sort

select[°] $\sqsubseteq$ **ok cons perm** $\implies \llbracket \text{nil}, \text{select}^\circ \rrbracket^\circ \sqsubseteq \text{perm ordered}$

(6.6.2a)

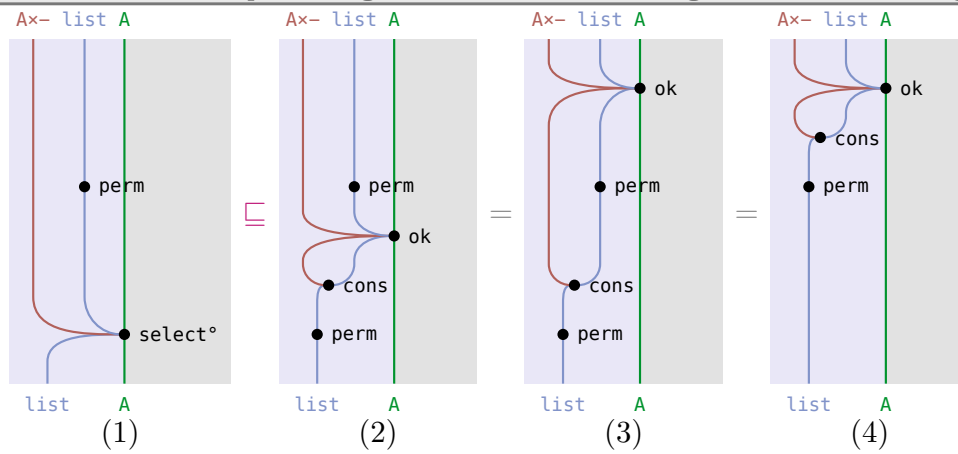
every output of unfolding the input by **select** is a sorted permutation of it



1	$\llbracket \text{nil}, \text{select}^\circ \rrbracket^\circ$	$\text{select}^\circ \sqsubseteq \text{ok cons perm}$
2	$\sqsubseteq (\llbracket \text{nil}, \text{ok cons} \rrbracket \text{ perm})^\circ$	$\implies \llbracket \text{nil}, \text{select}^\circ \rrbracket^\circ \sqsubseteq (\llbracket \text{nil}, \text{ok cons} \rrbracket \text{ perm})^\circ$
3	$= (\text{ordered perm})^\circ$	(6.6.1b) $\text{ordered} = \llbracket \text{nil}, \text{ok cons} \rrbracket$
4	$= \text{perm}^\circ \text{ ordered}^\circ$	$(R S)^\circ = S^\circ R^\circ$
5	$= \text{perm ordered}^\circ$	$\text{perm}^\circ = \text{perm}$
6	$= \text{perm ordered}$	$\text{ordered}^\circ = \text{ordered}$

$$\text{select}^\circ \sqsubseteq \text{ok cons perm} \Rightarrow (1 \times \text{perm}) \text{select}^\circ \sqsubseteq \text{ok cons perm} \quad (6.6.2b)$$

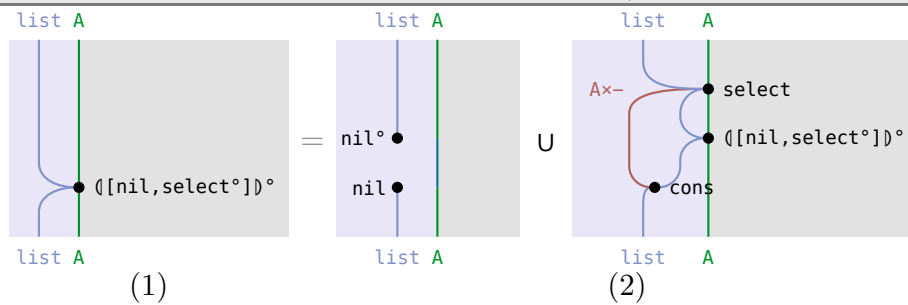
permuting the tail and then undoing **select** lands among the **ok** conses of a permutation



1	$(1 \times \text{perm}) \text{select}^\circ$	$\text{select}^\circ \sqsubseteq \text{ok cons perm}$
2	$\sqsubseteq (1 \times \text{perm}) \text{ok cons perm}$	$\text{ok } (1 \times \text{perm}) = (1 \times \text{perm}) \text{ok}$
3	$= \text{ok } (1 \times \text{perm}) \text{cons perm}$	$\text{cons perm} = (1 \times \text{perm}) \text{cons perm}$
4	$= \text{ok cons perm}$	

$$([nil, \text{select}^\circ])^\circ = (nil^\circ nil) \cup (\text{select } (1 \times ([nil, \text{select}^\circ])^\circ) \text{cons}) \quad (6.6.2c)$$

selection sort returns $[]$ on $[]$, and otherwise selects (a, y) , sorts y and conses a back on



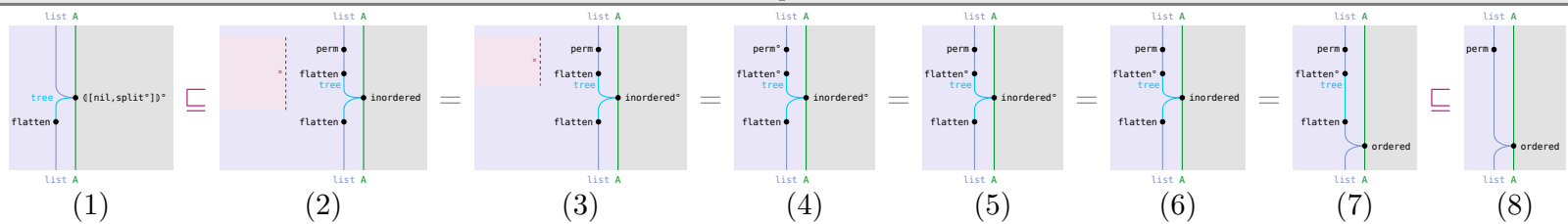
1	$[nil, \text{select}^\circ]^\circ$	unfolding the converse of a fold
2	$= (nil^\circ nil) \cup (\text{select } (1 \times ([nil, \text{select}^\circ])^\circ) \text{cons})$	

§6.6.3. Quicksort

$$(\forall a. \forall b. \forall c. R(a, b) \Rightarrow R(b, c) \Rightarrow R(a, c)) \wedge \text{split}^\circ \sqsubseteq \text{check}' \text{join perm} \Rightarrow ([nil, \text{split}^\circ])^\circ \text{flatten} \sqsubseteq \text{perm ordered} \quad (6.6.3a)$$

flatten $\sqsubseteq$ **perm ordered**

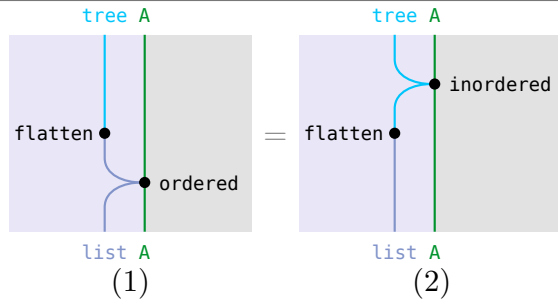
every output of unfolding the input by **split** into a tree and flattening that tree is a sorted permutation of the input



1	$[nil, \text{split}^\circ]^\circ \text{flatten}$	$\text{split}^\circ \sqsubseteq \text{check}' \text{join perm}$ $\Rightarrow ([nil, \text{split}^\circ])^\circ \text{flatten} \sqsubseteq (\text{inordered flatten perm})^\circ \text{flatten}$
2	$\sqsubseteq (\text{inordered flatten perm})^\circ \text{flatten}$	$(R S)^\circ = S^\circ R^\circ$
3	$= (\text{flatten perm})^\circ \text{inordered}^\circ \text{flatten}$	$(R S)^\circ = S^\circ R^\circ$
4	$= \text{perm}^\circ \text{flatten}^\circ \text{inordered}^\circ \text{flatten}$	$\text{perm}^\circ = \text{perm}$
5	$= \text{perm flatten}^\circ \text{inordered}^\circ \text{flatten}$	$\text{inordered}^\circ = \text{inordered}$
6	$= \text{perm flatten}^\circ \text{inordered flatten}$	$(6.6.3b) (\forall a. \forall b. \forall c. R(a, b) \Rightarrow R(b, c) \Rightarrow R(a, c))$ $\Rightarrow \text{flatten ordered} = \text{inordered flatten}$
7	$= \text{perm flatten}^\circ \text{flatten ordered}$	$\text{flatten}^\circ \text{flatten} \sqsubseteq 1$
8	$\sqsubseteq \text{perm ordered}$	

($\forall a. \forall b. \forall c. R(a,b) \Rightarrow R(b,c) \Rightarrow R(a,c)$) $\Rightarrow$ flatten ordered = inordered flatten (6.6.3b)

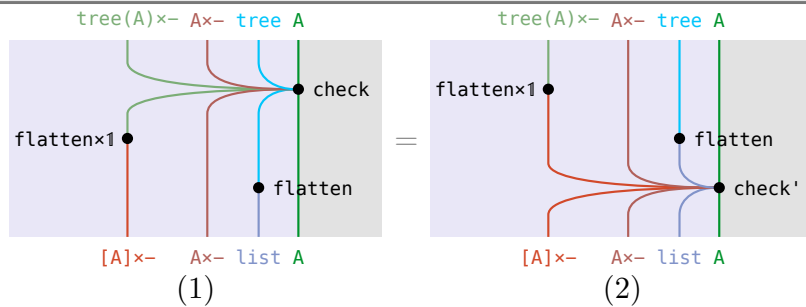
flattening a tree and then testing the list for order is the same as testing the tree with **inordered** and then flattening



1	flatten ordered	R transitive: a flattened tree is ordered iff every node passes check
2	= inordered flatten	

check (flatten $\times$ (1 $\times$ flatten)) = (flatten $\times$ (1 $\times$ flatten)) check' (6.6.3c)

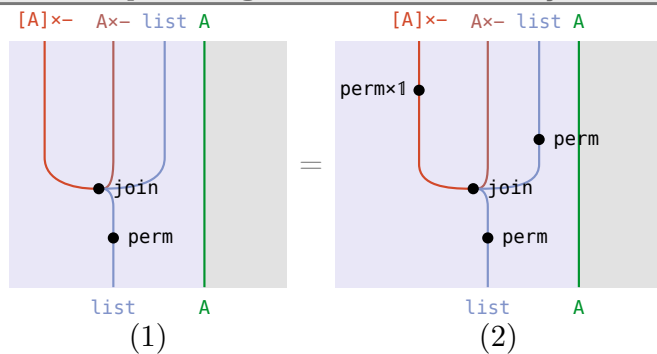
testing a fork with **check** and then flattening both subtrees equals flattening both subtrees and then testing with **check'**, since **intree** of a tree is **inlist** of its flattening



1	check (flatten $\times$ (1 $\times$ flatten))	b intree x $\Leftrightarrow$ b inlist flatten(x)
2	= (flatten $\times$ (1 $\times$ flatten)) check'	

join perm = (perm $\times$ (1 $\times$ perm)) join perm (6.6.3d)

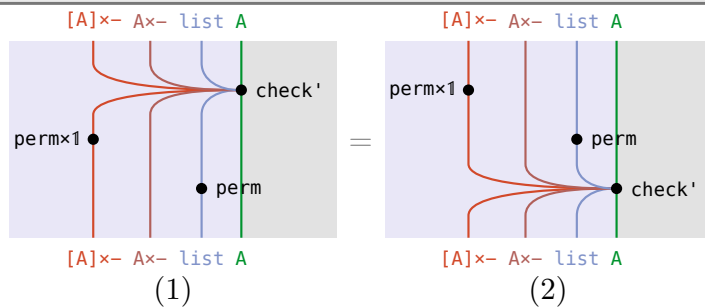
permuting the list **x ++ [a] ++ y** is the same as first permuting **x** and **y** and then permuting the joined list



1	join perm	perm is transitive and closed under ++
2	= (perm $\times$ (1 $\times$ perm)) join perm	

check' (perm $\times$ (1 $\times$ perm)) = (perm $\times$ (1 $\times$ perm)) check' (6.6.3e)

testing a fork of lists with **check'** and then permuting both parts equals permuting both parts and then testing, since permuting a list leaves its members unchanged

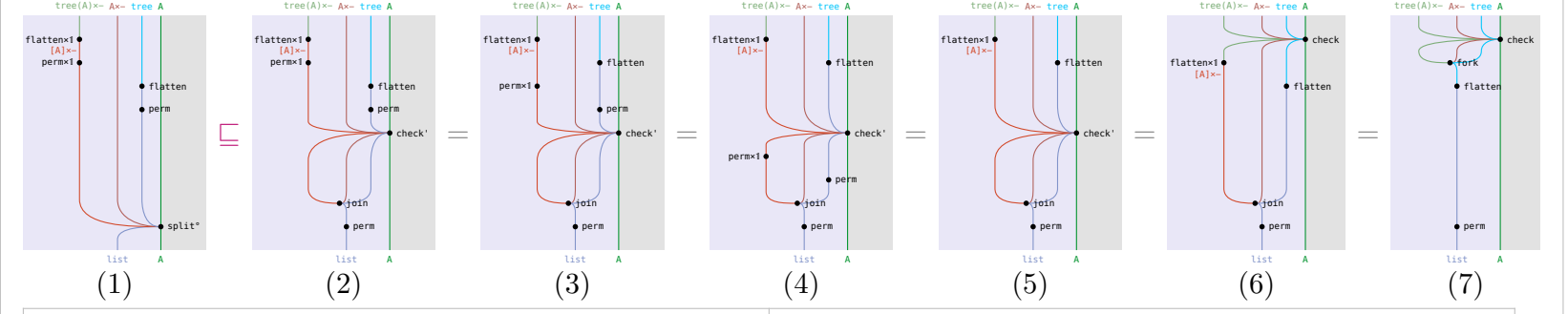


1	check' (perm $\times$ (1 $\times$ perm))	b inlist x $\Leftrightarrow$ b inlist perm(x)
2	= (perm $\times$ (1 $\times$ perm)) check'	

split° ⊆ check' join perm ⇒ ((flatten perm) × (1 × (flatten perm))) split° ⊆ check fork (6.6.3f)

flatten perm

undoing **split** and then flattening and permuting both parts lands among the **checked** forks whose flattening is **permuted**

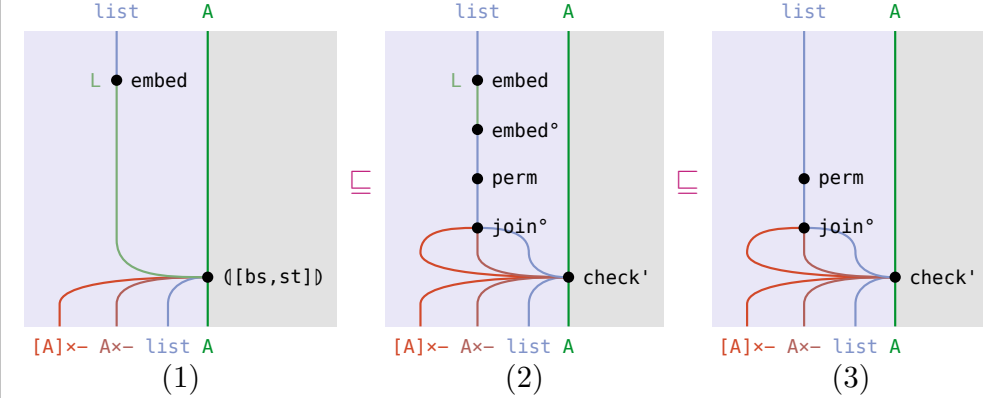


1	$((\text{flatten perm}) \times (1 \times (\text{flatten perm}))) \text{split}^\circ$	$\text{split}^\circ \sqsubseteq \text{check}' \text{ join perm}$
2	$\sqsubseteq ((\text{flatten perm}) \times (1 \times (\text{flatten perm}))) \text{check}' \text{ join perm}$	
3	$= (\text{flatten} \times (1 \times \text{flatten})) (\text{perm} \times (1 \times \text{perm})) \text{check}' \text{ join perm}$	(6.6.3e) $\text{check}' (\text{perm} \times (1 \times \text{perm})) = (\text{perm} \times (1 \times \text{perm})) \text{check}'$
4	$= (\text{flatten} \times (1 \times \text{flatten})) \text{check}' (\text{perm} \times (1 \times \text{perm})) \text{join perm}$	(6.6.3d) $\text{join perm} = (\text{perm} \times (1 \times \text{perm})) \text{join perm}$
5	$= (\text{flatten} \times (1 \times \text{flatten})) \text{check}' \text{ join perm}$	(6.6.3c) $\text{check} (\text{flatten} \times (1 \times \text{flatten})) = (\text{flatten} \times (1 \times \text{flatten})) \text{check}'$
6	$= \text{check} (\text{flatten} \times (1 \times \text{flatten})) \text{join perm}$	$\text{fork flatten} = (\text{flatten} \times (1 \times \text{flatten})) \text{join}$
7	$= \text{check fork flatten perm}$	

bs ⊆ wrap perm join° check' ∧ (1 × (perm join° check')) st ⊆ cons perm join° check' ⇒ (6.6.3g)

embed ([bs,st]) ⊆ perm join° check'

turning a list into a non-empty list and folding it with any **[bs, st]** meeting the two conditions below gives only **checked** triples **(x,a,y)** whose join $x \# [a] \# y$ is a permutation of the list

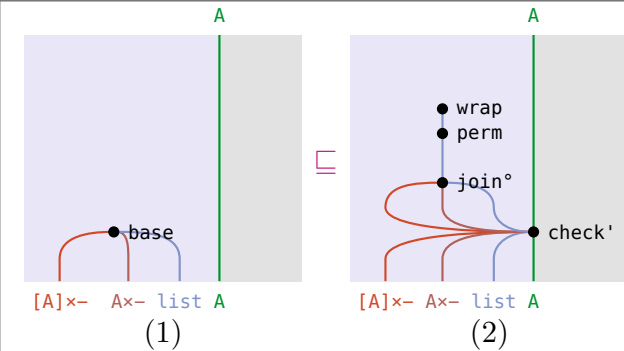


1	$\text{embed } ([bs, st])$	$\text{bs} \sqsubseteq \text{wrap perm join}^\circ \text{check}' \wedge (1 \times (\text{perm join}^\circ \text{check}')) \text{st} \sqsubseteq \text{cons perm join}^\circ \text{check}'$ $\Rightarrow \text{embed } ([bs, st]) \sqsubseteq \text{embed embed}^\circ \text{perm join}^\circ \text{check}'$
2	$\sqsubseteq \text{embed embed}^\circ \text{perm join}^\circ \text{check}'$	$\text{embed embed}^\circ \text{perm join}^\circ \text{check}' \sqsubseteq \text{perm join}^\circ \text{check}'$
3	$\sqsubseteq \text{perm join}^\circ \text{check}'$	

base ⊆ wrap perm join° check'

(6.6.3h)

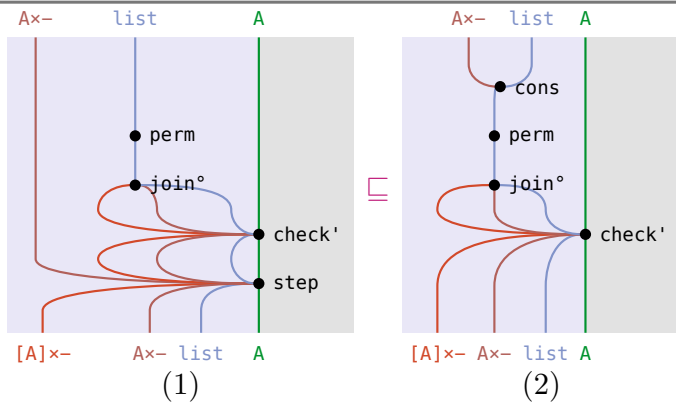
base(a) is a **checked** triple whose join is a permutation of the one-element list **[a]**



1	base	$[\] \# [a] \# [\] = [a]$, and check' holds on empty lists
2	$\sqsubseteq \text{wrap perm join}^\circ \text{check}'$	

connected(R) $\Rightarrow$ (1 $\times$ (perm join $^\circ$ check')) step $\sqsubseteq$ cons perm join $^\circ$ check' (6.6.3i)

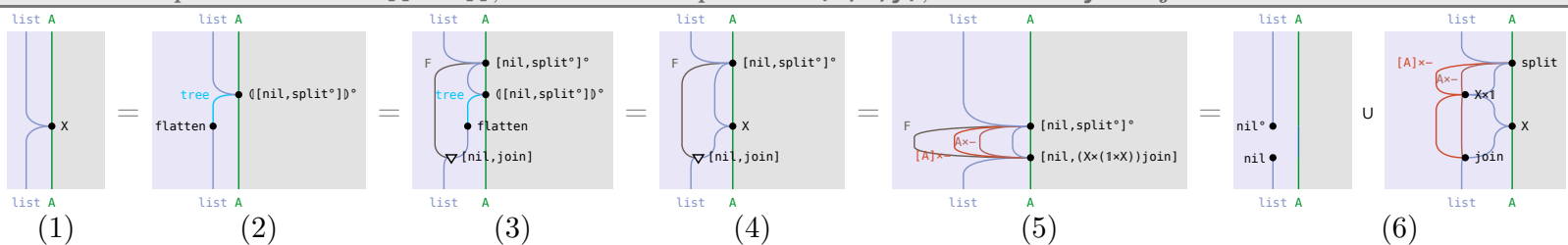
if (x,b,y) is a **checked** triple for a permutation of **l**, then **step(a, (x,b,y))** is a **checked** triple for a permutation of **[a] # l**



1	(1 $\times$ (perm join $^\circ$ check')) step	aRb puts a in front of x; otherwise bRa and a goes in front of y
2	$\sqsubseteq$ cons perm join $^\circ$ check'	

X = ([nil,split $^\circ$]) $^\circ$ flatten $\Rightarrow$ X = (nil $^\circ$ nil) u (split (X $\times$ (1 $\times$ X)) join) (6.6.3j)

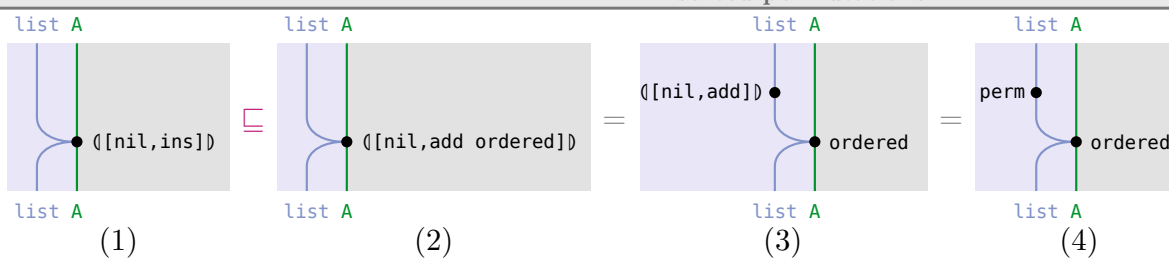
quicksort returns [] on [], and otherwise splits into (x,a,y), sorts x and y and joins them around a



1	X	X = ([nil,split $^\circ$]) $^\circ$ flatten
2	= ([nil,split $^\circ$]) $^\circ$ flatten	S $^\circ$ F((S) $^\circ$ (R)) R = (S) $^\circ$ (R)
3	= [nil,split $^\circ$] $^\circ$ F([nil,split $^\circ$] $^\circ$ flatten) [nil,join]	X = ([nil,split $^\circ$]) $^\circ$ flatten
4	= [nil,split $^\circ$] $^\circ$ F(X) [nil,join]	F(S) [T,U] = [T,(S $\times$ (1 $\times$ S)) U]
5	= [nil,split $^\circ$] $^\circ$ [nil,(X $\times$ (1 $\times$ X)) join]	[U,V] $^\circ$ [R,S] = (U $^\circ$ R) u (V $^\circ$ S)
6	= (nil $^\circ$ nil) u (split (X $\times$ (1 $\times$ X)) join)	

(1 $\times$ ordered) ins $\sqsubseteq$ add ordered $\Rightarrow$ ([nil,ins]) $\sqsubseteq$ perm ordered (6.6.3k)

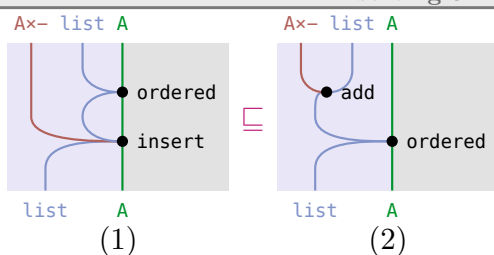
for any **ins** that, on an ordered list, returns only ordered results of **add**, folding with **[nil, ins]** gives only sorted permutations



1	[nil,ins]	(1 $\times$ ordered) ins $\sqsubseteq$ add ordered
2	$\sqsubseteq$ [nil,add ordered]	$\Rightarrow$ ([nil,ins]) $\sqsubseteq$ ([nil,add ordered])
3	= [nil,add] ordered	[nil,add] ordered = [nil,add ordered]
4	= perm ordered	perm = [nil,add]

connected(R) $\wedge$ ($\forall a. \forall b. \forall c. R(a,b) \Rightarrow R(b,c) \Rightarrow R(a,c)$) $\Rightarrow$ (1 $\times$ ordered) insert $\sqsubseteq$ add ordered (6.6.3l)

inserting **a** into an ordered list puts **a** somewhere in it and keeps it ordered



1	(1 $\times$ ordered) insert	insert splices a in, and R transitive and connected
2	$\sqsubseteq$ add ordered	

§6.7. Closure

$$R^* = (\mu X : 1 \cup (X R)) \quad (6.7a)$$

closing R by composing it on the left and closing it by composing on the right give the same relation R^*

$$1 \in X \wedge X X \subseteq X \implies R \subseteq X \iff R^* \subseteq X \quad (6.7b)$$

X contains R exactly when it contains R^*

$$1 \in R^* \quad (6.7c)$$

R^* is reflexive

1	1	$R \subseteq R \cup S$
2	$\sqsubseteq 1 \cup (R R^*)$	$1 \cup (R R^*) = R^*$
3	$= R^*$	

$$R \subseteq R^* \quad (6.7d)$$

R^* contains R

1	R	
2	$= R$	$(6.7c) 1 \in R^*$
3	$\sqsubseteq R R^*$	$R R^* \subseteq R^*$
4	$\sqsubseteq R^*$	

$$(1 \cup (R (R^* / R^*))) R^* \subseteq R^* \quad (6.7e)$$

R^*/R^* is a prefixed point of $X \mapsto 1 \cup R X$, stated through division

$$R^* R^* \subseteq R^*$$

R^* is transitive, because R^*/R^* is a prefixed point of $X \mapsto 1 \cup R X$

1	$(1 \cup (R (R^* / R^*))) R^*$	$(S \cup T) R = (S R) \cup (T R)$
2	$= (R^*) \cup (R (R^* / R^*) R^*)$	
3	$= R^* \cup (R (R^* / R^*) R^*)$	$(R / R) R \subseteq R$
4	$\sqsubseteq R^* \cup (R R^*)$	$R R^* \subseteq R^*$
5	$\sqsubseteq R^*$	

$$1 \in X \wedge X X \subseteq X \wedge R \subseteq X \implies 1 \cup (R X) \subseteq X \quad (6.7f)$$

a reflexive transitive X containing R is a prefixed point of $Y \mapsto 1 \cup R Y$

$$1 \in X \wedge X X \subseteq X \wedge R \subseteq X \implies R^* \subseteq X$$

a reflexive transitive X containing R is a prefixed point of $X \mapsto 1 \cup R X$, so it contains R^*

1	$1 \cup (R X)$	$1 \in X$
2	$\sqsubseteq X \cup (R X)$	$R \subseteq X$
3	$\sqsubseteq X \cup (X X)$	
4	$\sqsubseteq X$	

$$\frac{R^*}{\exists} = \left(\frac{1}{\exists}, \frac{R R^*}{\exists} \right) \text{ cup} \quad (6.7g)$$

the set of R^* -successors of a is a itself joined with the R^* -successors of its R -successors

1	$\frac{R^*}{\exists}$	$1 \cup (R R^*) = R^*$
2	$= \frac{1 \cup (R R^*)}{\exists}$	$\frac{R \cup S}{\exists} = \left(\frac{R}{\exists}, \frac{S}{\exists} \right) \text{ cup}$
3	$= \left(\frac{1}{\exists}, \frac{R R^*}{\exists} \right) \text{ cup}$	

§6.7.1. Computing closure

$R - \emptyset = R$	(6.7.1a)
taking nothing away leaves R	
$R \cup (S - R) = R \cup S$	
beside R , only the part of S outside R adds anything	
$R - (S \cup T) = R - S - T$	
taking away S and then T takes away $S \cup T$	
$(R \cup S) - T = (R - T) \cup (S - T)$	
taking T away from a union takes it away from each part	

$\text{monotonic}(\varphi) \wedge \text{monotonic}(\psi) \implies (\mu X : \varphi(\psi(X))) = \varphi(\mu Y : \psi(\varphi(Y)))$	(6.7.1b)
the least fixed point of φ after ψ is φ applied to the least fixed point of ψ after φ	

$S R^* = (\mu X : S \cup (X R))$	(6.7.1c)
S followed by any number of R steps is the least X containing S and closed under a further R step	
$R^* S = (\mu X : S \cup (R X))$	
any number of R steps followed by S is the least X containing S and closed under an R step in front	

$\theta(\emptyset, S) = S R^*$	(6.7.1d)
started with nothing found and S to explore, θ computes SR^*	
1 $\theta(\emptyset, S)$	$\theta(P, Q) \triangleq P \cup (\mu X : Q \cup ((X R) - P))$
2 $\triangleq \emptyset \cup (\mu X : S \cup ((X R) - \emptyset))$	(6.7.1a) $R - \emptyset = R$
3 $= \emptyset \cup (\mu X : S \cup (X R))$	$\emptyset \cup R = R$
4 $= (\mu X : S \cup (X R))$	(6.7.1c) $S R^* = (\mu X : S \cup (X R))$
5 $= S R^*$	

$\theta(P, \emptyset) = P$	(6.7.1e)
with nothing left to explore, θ returns what it has found	
1 $\theta(P, \emptyset)$	$\theta(P, Q) \triangleq P \cup (\mu X : Q \cup ((X R) - P))$
2 $\triangleq P \cup (\mu X : \emptyset \cup ((X R) - P))$	$(\mu X : \emptyset \cup ((X R) - P)) = \emptyset$
3 $= P \cup \emptyset$	$R \cup \emptyset = R$
4 $= P$	